Project 2: Image Stitching

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May 2, 2025

1 Description

This project aims to stitch a set of images into a seamless panorama. The process involves several key image processing steps, including feature detection, description, and matching. Once corresponding features are matched across images, further operations such as warping, blending, alignment and cropping are performed to generate the final panoramic output. Image stitching enhances both the effective resolution and the field of view, making it a powerful technique for producing wide-angle visual representations.

2 Processing Flow

1. Feature Detection
    - Moravec Corner Detector
    - Harris Corner Detector (Bonus)
2. Feature Description
3. Feature Matching
4. Image matching
5. Blending
6. End-to-end Alignment (Bonus)
7. Rectangling (Bonus)

3 Feature Detection

3.1 Moravec Corner Detector

We first implement the simple Moravec corner detector. For each pixel, the algorithm calculates the intensity difference between its patch and a shifted version. Then, it uses the minimum difference among four shifted directions as the pixel's energy. A corner exhibits drastic intensity changes in all directions, and thus results in higher energy.

$$E(\mathbf{u}) = \sum_{X} w(\mathbf{x})[I(\mathbf{x}+\mathbf{u}) - I(\mathbf{x})]^2$$

3.2 Harris Corner Detector

However, Moravec has several problems. First, it uses a binary window function for the patch, which makes it more susceptible to noise. Second, it only considers shifts in eight directions, potentially missing corners at other angles.

The Harris corner detector introduces an alternative method for computing the energy to address these issues. It uses a Gaussian function to produce a smoother window for the patch. Then, it applies Taylor expansion to approximate the intensity difference for a given shift direction, which can be expressed as:

$$E(\mathbf{u}) \approxeq \mathbf{u}^\top \mathbf{M} \mathbf{u}, \quad \mathbf{M} = \sum_X w(\mathbf{x}) \begin{bmatrix} I_x^2 & I_x I_y \\ I_y^2 & I_y^2 \end{bmatrix}$$

Since E is a quadratic function of the shift vector $\mathbf{u}$, we can analyze its shape by examining the properties of $\mathbf{M}$. As a result, Harris considered the eigenvalues of $\mathbf{M}$. When both eigenvalues are large, it indicates that E will be large in all directions $\mathbf{u}$, suggesting the presence of a corner. Harris proposed the following energy function:

$$E = \det\mathbf{M} - k(\mathrm{tr}\mathbf{M})^2 = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$$

3.3 Comparison

In Figure 1, we demonstrate the results of the Moravec and Harris algorithms. In all the images, redder colors indicate higher values, while bluer colors indicate lower values.

Both algorithms performs well in identifying the corners or possible feature locations. In the Harris algorithm column, we observe that the horizontal and vertical lines appear in blue, which aligns with our expectations. However, we did not anticipate that a diagonal line would appear in red and be classified as a corner.

3.4 Threshold

3.5 Maximal Suppress

3.6 Interactive Gadget

4 Feature Description

5 Feature Matching

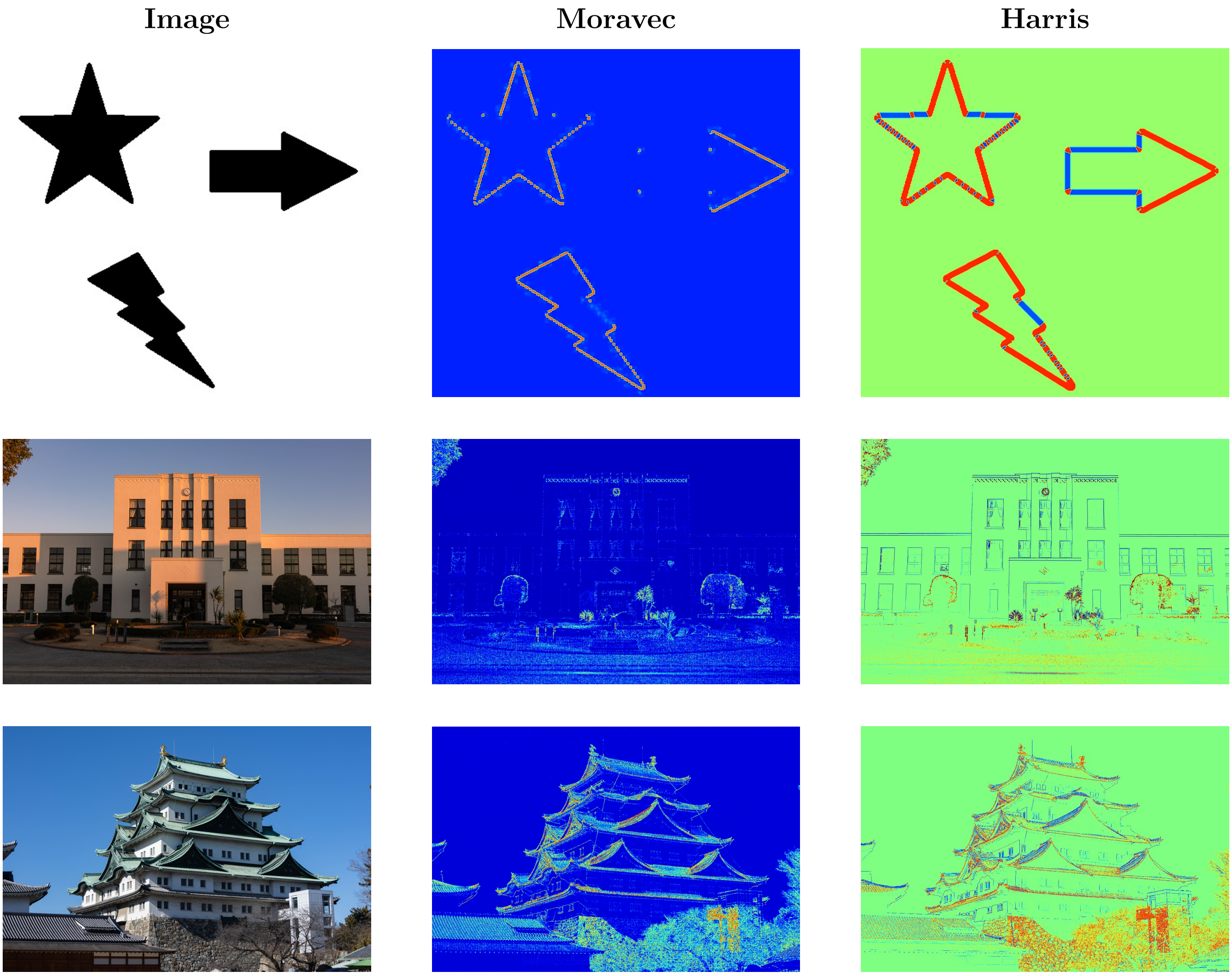

Figure 1: Comparison of Moravec and Harris Algorithms